

Agricultural Flood Risk Assessment Model

Technical Documentation of R Implementation

This document describes an R-based modelling workflow used to estimate agricultural flood risk under multiple climate scenarios. The script integrates flood depth outputs from hydrological simulations with crop yield and price data to calculate expected economic damage. Flood impacts are estimated using Joint Research Centre (JRC) depth–damage functions, which convert flood depth into a proportion of crop damage. These damage estimates are combined with exposure metrics such as cultivated area, crop yield, and market price to estimate financial risk across different climate scenarios.

Dataset Description

The script uses several datasets representing exposure, hazard, and vulnerability components of the climate risk framework. Input data include: Hazard data: • Flood depth outputs derived from extreme precipitation modelling • Scenario-specific flood datasets including: - Historical 100-year flood event - RCP 2.6 100-year flood event - RCP 4.5 100-year flood event - RCP 8.5 100-year flood event - RCP 8.5 1000-year flood event Exposure data: • Crop price data (€/tonne) • Crop yield statistics (tonnes per square meter) Vulnerability data: • JRC flood depth–damage curves describing the percentage crop damage as a function of flood depth.

Climate Risk Modelling Framework

The modelling workflow follows a simplified climate risk framework: $\text{Risk} = \text{Hazard} \times \text{Exposure} \times \text{Vulnerability}$ Where: Hazard • Flood depth estimates produced from hydrological modelling Exposure • Agricultural area exposed to flooding • Crop yield and crop price Vulnerability • JRC damage functions translating flood depth into expected damage fraction The script combines these elements to estimate economic losses for each scenario.

1. Library Dependencies

The following R packages are used: • dplyr – data manipulation and transformation • ggplot2 – visualization and exploratory analysis • scales – formatting numeric values for plots These packages support the data processing and reporting stages of the workflow.

2. Data Loading

Multiple datasets are imported using `read.csv()`. These include: • `crop_prices.csv` – crop market prices • `YPH_statsline.csv` – crop yield statistics • `jrc_dd_functions.csv` – depth–damage curves • flood scenario datasets – flood depths associated with different climate scenarios These datasets are used to construct the hazard, exposure, and vulnerability components of the risk calculation.

3. JRC Damage Function

A custom function is defined to calculate flood damage percentages based on flood depth values. The function uses linear interpolation via `approx()`: `calculate_jrc_damage(depth_vector, dd_df)` This interpolates damage percentages from the JRC depth–damage curve for the given flood depth values. If flood depths fall outside the available range, extrapolation rules ensure the nearest valid damage value is applied.

4. Scenario Processing Function

The core analysis is implemented in the function: `process_scenario(df, scenario_name)` This function processes each flood scenario dataset and performs the following steps: 1. Attach crop price and yield data 2. Filter flood depths to valid ranges 3. Calculate flood damage using the JRC damage function 4. Estimate exposure metrics such as cultivated area 5. Compute economic risk values 6. Assign scenario labels for downstream analysis

5. Flood Depth Filtering

Flood depth values are constrained to realistic bounds. If depth values exceed the maximum supported by the JRC damage curves, they are capped at the maximum value. Similarly, negative flood depths are replaced with zero. This prevents unrealistic damage calculations and ensures compatibility with the vulnerability functions.

6. Vulnerability Calculation

The vulnerability component represents the fraction of crop damage caused by flooding. Damage values are derived from the JRC depth–damage curves, which express damage as a percentage between 0 and 100%. These values represent the proportion of crop yield expected to be lost at a given flood depth.

7. Exposure Calculation

Exposure represents the economic value of agricultural production that could be affected by flooding. Exposure is calculated as: $\text{Exposure} = \text{Area} \times \text{Crop Yield} \times \text{Crop Price}$ Where: • Area is derived from the number of raster pixels representing farmland • Each pixel corresponds to a 4 m² grid cell • Yield and price data convert the exposed area into economic value.

8. Risk Calculation

Economic flood risk is calculated as: $\text{Risk} = \text{Probability} \times \text{Exposure} \times \text{Vulnerability}$ In this model: • Probability corresponds to the return period of the flood event • Exposure represents crop production value • Vulnerability is the damage fraction derived from the JRC curves The script calculates both: • Total scenario risk • Risk per individual agricultural plot

9. Scenario Iteration

The script loops through all flood scenarios using a named list of datasets. For each scenario: • Flood depths are processed • Damage functions are applied • Risk values are calculated • Scenario results are stored in structured lists

10. Results Aggregation

Scenario outputs are aggregated into combined result tables. Two major datasets are produced: • `final_results` – detailed plot-level risk calculations • `final_summary` – aggregated results by crop type and scenario Aggregation includes metrics such as: • Total economic risk • Average flood depth • Mean damage ratio • Number of agricultural plots • Total affected agricultural area.

11. Crop Name Standardization

Crop names in the source datasets appear in Dutch. To enable consistent analysis, crop names are mapped to English equivalents. Examples include: • Aardappel → Potatoes • Bieten (suiker) → Sugar Beets • Tarwe → Wheat This step ensures consistent grouping during summary calculations.

12. Scenario Risk Comparison

The script calculates percentage increases in risk relative to historical baseline conditions. For each crop type: $\text{Percentage Increase} = (\text{Current Scenario Risk} - \text{Historical Risk}) / \text{Historical Risk}$ This provides a simple metric to quantify how climate change may increase agricultural flood risk.

13. Application in Climate Risk Assessment

The modelling framework can support several climate risk applications: • Agricultural climate risk analysis • Flood risk assessments for rural regions • Economic stress testing under climate scenarios • Climate adaptation planning for crop systems By combining hazard modelling with economic exposure data, the workflow produces scenario-based estimates of financial losses due to extreme flooding.